

ON THE OPPORTUNITIES OF GREEN COMPUTING: A SURVEY

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ABSTRACT

Artificial Intelligence (AI) has achieved significant advancements in technology and research with the development over several decades, and is widely used in many areas including computing vision, natural language processing, time-series analysis, speech synthesis, etc. During the age of deep learning, especially with the arise of Large Language Models, a large majority of researchers' attention is paid on pursuing new state-of-the-art (SOTA) results, resulting in ever increasing of model size and computational complexity. The needs for high computing power brings higher carbon emission and undermines research fairness by preventing small or medium-sized research institutions and companies with limited funding in participating in research. To tackle the challenges of computing resources and environmental impact of AI, Green Computing has become a hot research topic. In this survey, we give a systematic overview of the technologies used in Green Computing. We propose the framework of Green Computing and devide it into four key components: (1) Measures of Greenness, (2) Energy-Efficient AI, (3) Energy-Efficient Computing Systems and (4) AI Use Cases for Sustainability. For each components, we discuss the research progress made and the commonly used techniques to optimize the AI efficiency. We conclude that this new research direction has the potential to address the conflicts between resource constraints and AI development. We encourage more researchers to put attention on this direction and make AI more environmental friendly.

Keywords Artificial Intelligence · Green Computing · Carbon Footprint · AI Sustainability